

Chest Pain: Suspected and Confirmed Acute Coronary Syndrome Management Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All Clinical Services	Nursing and Medical

2. Purpose

To mandate the implementation of minimum practice standards for the evaluation and management of Chest Pain and Acute Coronary Syndrome (ACS) of the adult hospitalised patient throughout Armadale Kalamunda Group (AKG).

Mandatory Requirements Emergency Department

- On initial presentation to the Emergency Department (ED) the patient with chest pain is to be assigned a triage score of 1 or 2 by a triage competent nurse. The patient will be allocated to cardiac monitored area. All patients presenting with chest pain must be triaged according to the National Australasian Triage Scale on assessment.
- All patient's with Chest pain suspecting of ACS should be seen in the resuscitation bay
- A 12-Lead ECG is to be performed within 10 minutes of arrival and reviewed by a Registrar. It should be repeated at the onset of each episode of chest pain/ischaemic symptom(s) and repeated as clinically appropriate or as directed by MO.
- Provide Supplemental oxygen to aim a SaO₂ of > 94%; or 88-92% in the presence of COPD or Type II Respiratory Failure as directed by the ED MO
- Ascertain and document any pre-hospital treatments to patients.
- If patient is identified as a STEMI or a high risk NSTEMI/ACS, prepare for Coronary Care Transfer (CCT), Coronary Care Unit (CCU) or appropriate site-specific department admission as requested by ED/CCU physician. Upon transfer, the patient must be attached to a defibrillator, have O₂, suction, air-viva and resuscitation medications available. Transfers do not need a doctor and / or nurse, as paramedics have the required skills for most inter-hospital transfers.

Mandatory Requirements Ward/Unit

- Perform a 12 lead ECG and have an MO review within 10 minutes from onset of pain
- Document on ECG:
 - Presence of chest pain and nature
 - Pain score (1-10)
 - Date and time (check automatic print out is correct). May be inaccurate if the machine is not correctly programmed

- Label ECG with patient name, date of birth +/- unit medical record number (UMRN)
 - Signed by medical staff on review
- Attach to cardiac monitoring as clinically indicated
- Complete initial assessment as outlined below
- Escalate to cardiology as appropriate
- Call a MET if clinically required

3. Procedure

3.1 General information

- The clinical presentation of myocardial ischemia is most often acute chest discomfort. The goal of emergency department and wider hospital management evaluation is to determine the cause of the chest pain and promptly initiate appropriate therapy. It is essential that initial assessment and management be rapid but methodical and evidence based.
- If untreated, the prognosis is poor and mortality high, particularly in people who have had myocardial damage. Appropriate triage, risk assessment and timely use of acute pharmacological or invasive interventions are critical for the prevention of future adverse cardiovascular events (myocardial infarction, stroke, repeat revascularisation or death).
- All patients complaining of chest pain/discomfort should be suspected of having ACS until proven otherwise.

3.2 Initial assessment of ACS chest pain

Initial assessment and management of ACS Chest Pain should aim to achieve the following:

- Facilitate immediate management of care upon first emergency clinical contact.
- Obtain 12-Lead ECG with interpretation by an MO within 10 minutes of ED arrival.
- Rapid assessment of Airway, Breathing, Circulation and Pain with documented baseline observations
- Timely reperfusion for patients with STEMI, in accordance with time frames recommended by current guidelines
- Timely administration of pharmacological interventions
- Risk stratification and management of patients with a NSTEMI/ACS based on a documented, evidence-based assessment of their risk for an adverse event.

3.3 Facilitate immediate management of care

- All patients presenting with onset of chest pain or other symptoms suggestive of myocardial ischemia (e.g. sweating, sudden orthopnoea, dyspnoea, syncope, epigastric discomfort, jaw pain, or arm pain) within the last 48 hours should be assigned Triage category 2 and brought immediately into a cardiac monitored bay.
- The Australasian Triage Scale (ATS) recommendation for patients presenting to the ED with chest pain is to commence assessment within 10 minutes of presentation.
- Consideration should be given to patient cohorts in whom atypical presentations of ACS are more frequently encountered (e.g. people with diabetes, women, older patients, those with mental illness, those from culturally and linguistically diverse (CALD) populations and Aboriginal and Torres Strait Islander peoples).

3.4 12-lead ECG analysis within 10 minutes of presentation/notification of chest pain

- Obtain 12-Lead ECG within ten minutes of ED presentation or onset of chest pain.
- Ensure ECG is reviewed and interpreted by a Medical Officer (MO) within 10 minutes (as practicable) from onset of pain or presentation to the ED. Interpretation should be performed by a clinician with expertise in ECG analysis. In the ED, only a medical officer of registrar level or above should sign off ECGs.
- Ensure ECG is labelled with correct patient name, date and time of ECG recording, and pain score.
- The ECG is a crucial component of risk assessment and planning of treatment. It provides information about the site and extent of myocardial ischemia or damage and in determining whether the patient may need emergency reperfusion (thrombolytic therapy or percutaneous coronary intervention).
- Where ECG changes are indicative of STEMI, immediate referral to cardiology with regard to patient transfer to tertiary facility for Primary Clinical Intervention (PCI). Refer to STEMI guidelines.
- Serial ECGs should be taken every 10-15 minutes until the patient is pain free and compared in sequence, and where possible, with pre-existing ECGs.

3.5 Rapid assessment of Airway, Breathing, Circulation

- Baseline observations are important to identify changes in the patient's condition and to assess cardiac function.
- A systematic approach to obtaining baseline observations may be helpful in ensuring thorough initial assessment is performed.

	Evaluate and document	Interventions / Actions
Airway	Assess patency	Secure as clinically indicated
Breathing	• Respiratory rate (RR) • Signs of dyspnoea, effort • Peripheral oxygen saturation (SpO₂)	• Administer supplemental O₂ in hypoxic/shocked patients • Target O₂ saturations $\geq$ 94%; 88-92% in pts with COPD, Type II respiratory failure. Avoid hyperoxia and Hypoxia in ACS suspect patients
Circulation	• Heart rate (HR) • Heart rhythm e.g. regular, irregular • Pulse quality e.g. full, thready, bounding • Bilateral blood pressures (BP) • Presence and quality of pedal pulses • Signs of peripheral oedema	• Commence continuous cardiac monitoring (CCM) • 5-10 minutely vital signs – dependent on clinical presentation • If difference in Systolic BP between the both arms are more than 20 mm of Hg then aortic dissection should be suspected • IV access obtain bloods for laboratory testing (FBC, UEC,

	General appearance e.g. skin colour, diaphoresis	Troponin, +/- coagulation profile) Check with MO for further guidance
Disability	- Blood sugar level (BSL) - Glasgow Coma Scale (GCS) - Pain assessment	- Administer sublingual GTN for pain/hypertension. Watch for hypotension, Inferior wall STEMI and Right ventricular STEMI before administering the GTN - Aspirin 300mg chewed or dissolved - Titrated morphine/fentanyl for management of pain
Environment Exposure	Temperature	Undress patient and place in hospital gown for secondary assessment

- Perform manual blood pressure and pulse in order to obtain accurate pulse information, e.g. rate, depth, volume, bounding, regularity
- Assess nature and location of pain using the L.O.C.A.T.E criteria may be useful for assessment – Refer to [Appendix I](#): Chest Pain Tool – L.O.C.A.T.E
 - Location of pain
 - Onset and Duration
 - Character
 - Associated Symptoms
 - Treatment
 - Eliminate
- Ascertain the following from pain assessment:
 - Onset of pain and / or symptoms, duration of pain.
 - Location of pain, radiation to other body parts.
 - Quality of pain e.g. sharp, heavy, dull, crushing, tearing.
 - Pain score 0-10.
 - Ascertain precipitating factors e.g. anxiety, physical exertion, rest, change in haemodynamic parameters such as atrial fibrillation.
 - Assess whether pain is reproducible on deep inspiration, positional or with chest wall palpation
 - Note the presence of accompanying symptoms e.g. diaphoresis, pallor, tachycardia.
 - Comparison to previous episodes of chest pain (if applicable).
 - Factors which increase myocardial oxygen demand, can precipitate ischaemic chest pain.
- Duration of chest pain is important in determining the risk stratification for the patient involved.
 - Prolonged or unrelieved chest pain may be indicative of a myocardial infarction rather than unstable angina.

- In some cases, the patient cannot quantify the nature of the discomfort but places his or her fist in the centre of the chest (the “Levine sign”).
- Immediately report tearing, crushing or ripping type pain to the MO

3.6 Timely reperfusion for patients with STEMI

- The 12-Lead ECG along with patient history, examination and cardiac enzyme biomarkers e.g. cardiac specific troponin, provides information about the site and extent of myocardial ischemia or damage. Identification of patients with STEMI is essential in determining whether the patient may need emergency reperfusion, either:
- Thrombolytic therapy to dissolve the occluding thrombus, or
- Percutaneous Coronary Intervention (PCI) to open a blocked coronary artery.
- If delivered within the first few hours after onset of chest pain, reperfusion therapy and provide reduction in necrosis of cardiac tissue (myonecrosis), which in turn significantly improves prognosis.
- Individuals with NSTEMI/UA do not require emergency reperfusion but risk must be assessed to determine the most appropriate treatment and its timing.

3.7 Timely administration of pharmacological interventions

3.7.1 GTN

- In the presence of ongoing chest pain, nitro-glycerine (GTN) sublingual tablet (0.3-0.6mg) or spray (0.4-0.8mg) should be administered every five minutes for up to three doses if no contraindications exist (such as hypotension).
- Beneficial haemodynamic effects of GTN include improved myocardial oxygen delivery and reduction in myocardial demand.
 - Therapeutic effect is apparent within 1-2 minutes of IV administration.
 - Caution is advised in patients with aortic stenosis, inferior wall STEMI, right ventricular STEMI or high sensitivity to this drug.
- **All patients** should be questioned about the use of phosphodiesterase-5 inhibitors, such as sildenafil (viagra), vardenafil (levitra), or tadalafil (cialis); nitrates are contraindicated if these drugs have been used within the last 24 hours because of the propensity to cause potentially severe hypotension.

3.7.2 Aspirin

- In all patients with suspected ACS and without contraindications, aspirin (300mg orally, dissolved or chewed) should be administered as soon as possible after presentation.
- The administration of aspirin as an anti-platelet agent, given early, acts to reduce mortality in acute myocardial infarction and to prevent re-infarction.
- Additional anti-platelet and anticoagulation therapy should not be given to patients without a confirmed or probable diagnosis of ACS.

3.7.3 Morphine/Fentanyl

- In the absence of contraindications, it is reasonable to administer titrated morphine or fentanyl intravenously for ongoing chest discomfort at any time during the initial management.
- Caution should be taken to avoid sedation and respiratory depression.

- **Note:** morphine administration has been shown to delay gastric emptying and slow absorption of oral ACS medications, in particular oral P2y12 inhibitors, such as clopidogrel or ticagrelor.
 - This should be taken into consideration when planning peri-procedural anti-platelet therapies.

3.7.4 Oxygen supplementation

- Supplement oxygen therapy, unless contraindicated, to achieve a SaO₂ of > 94%.
- Routine use of oxygen supplementation for patients with normal blood oxygen saturation (94-98%) is not recommended.
 - Hyperoxia is known to cause coronary vasoconstriction and may reduce coronary blood flow.
 - Therefore, in the absence of hypoxia, the administration of oxygen therapy may cause harm.
- Nurse initiated supplemental oxygen is to be documented in the patient medical record.
 - Ongoing oxygen therapy must be prescribed on a dedicated oxygen prescription form by a medical officer.

4. Early risk stratification and management of NSTEMI/ACS patients

- The process of risk stratification is to assist in estimating the probability of ACS and ACS-related morbidity and mortality.

Recommended Further Management		
High risk Admit or Transfer	Intermediate Risk Rest ratify	Low Risk Discharge
☐ Continuous cardiac monitoring & frequent vital signs ☐ Repeat ECG immediately if symptoms recur ☐ Repeat ECG 2 hours post onset of symptoms ☐ Repeat Troponin at 2 hours if 1st sample negative ☐ Repeat ECG 6 hours post onset of symptoms ☐ Repeat Troponin at 6 hours Antiplatelet therapy ☐ 300mg Aspirin, dissolved or chewed ☐ 180mg Ticagrelor PO unless contraindicated ☐ 300mg or 600mg Clopidogrel PO Anticoagulant ☐ S/C Enoxaparin ☐ IV Heparin	☐ Continuous cardiac monitoring & frequent vital signs ☐ Repeat ECG immediately if symptoms recur ☐ Repeat ECG 2 hours post onset of symptoms ☐ Repeat Troponin at 2 hours if 1st sample negative ☐ Repeat ECG 6 hours post onset of symptoms ☐ Repeat Troponin at 6 hours ☐ ECG/Troponin review by MO Refer for Exercise Stress Test if: ☐ No further chest pain/symptoms and ☐ 2 negative troponin tests and ☐ No new ECG changes and ☐ No contraindications to stress test Re-stratify to High Risk if: ☐ Recurrent ischaemic chest pain or	☐ Regular vital signs ☐ Repeat ECG immediately if symptoms recur ☐ Repeat ECG 2 hours post onset of symptoms ☐ Repeat troponin at 2 hours if 1st sample negative ☐ ECG/Troponin review by medical officer Re-stratify Risk if: ☐ Recurrent ischaemic chest pain or ☐ Positive troponin or ☐ New ECG changes If low risk ACS ☐ Discharge ☐ Follow up GP within 3 days of discharge ☐ Consider specialist follow up

Symptomatic treatment of ongoing pain/hypertension ☐ IV GTN (titrate against pain & BP) ☐ IV morphine ☐ Refer to nominated cardiologist for further management	☐ Positive troponin or ☐ New ECG changes or ☐ Positive stress test Re-stratify to Low risk & discharge if: ☐ Negative stress test or ☐ Stress test available within 72 hours** and ☐ No further chest pain symptoms ☐ Repeat ECG and vital signs, if stable – discharge NB:** If stress test is not available within 72 hours of discharge, treatment plan should be guided by nominated SMO/Cardiologist Pharmacological stress test or CT coronary angiography may be indicated	☐ Consider discharge on aspirin (discuss with SMO) ☐ Vital signs prior to discharge If unlikely cardiac cause Consider alternative diagnosis Exit pathway
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- In patients presenting acutely with chest pain, this process aids evaluation, treatment (drug therapies or an early invasive therapeutic approach) and disposition (cardiac care unit, monitored environment, short stay units or discharge).

Table adapted from NSW Government Chest Pain Pathway.

http://www1.health.nsw.gov.au/pds/ActivePDSDocuments/PD2011_037.pdf

4.1 Cardiac biomarkers

- Cardiac troponins are the most sensitive and specific biomarker for myocardial injury and necrosis.
 - Both troponin I and T subtypes are cardio specific.
 - Troponin levels become elevated in the blood stream within 1-3 hours after AMI and may remain elevated for up to 14 days.
 - A high specific Troponin I (ortho-hs) serial testing is done at presentation, 2 hours and 4 hours to identify NSTEMI/ACS

4.2 Patient education

- Provide reassurance and rationale for all care provided.
 - Informing patient of care may reduce anxiety and therefore reduce the workload of the heart.
- Ensure call bell is within patient reach.
- Ask patient to remain resting in bed to decrease the risk of further ischaemic pain from increased myocardial oxygen demand.
- Ask patient to inform nursing staff immediately of any chest pain or other symptoms e.g. heaviness, pressure, arm pain etc.
 - Prompt reporting enables early treatment of chest pain.
- Explain to the patient nature of illness and ways to reduce further incidents (e.g. lifestyle advice).
 - Advice of lifestyle changes will reduce the risk of further coronary heart disease events.

4.3 Common Pharmacological Interventions in the Management of ACS Chest pain

- Nitrate therapy
- Anti-Platelet therapy
- Anti-Thrombin therapy
- Beta Blocker therapy
- Opioid analgesia
- ACE inhibitors
- Lipid Lowering therapy

5. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- HDWA [Consent to Treatment Policy OD 0657-16](#)
- HDWA [High Risk Medication Policy MP 0131-20](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service [Inter-Hospital Patient Transfer Policy EMHS: 38](#)

6. Supporting information

The following documents support the implementation of this procedure:

- [Clinical Handover Procedure AKG-PRO-0521-19](#)
- [Electrocardiogram 12 Lead Recording and Review Procedure AKG-PRO-0145-11](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)
- [Patient Transfer and Escort Guideline AKG-GUI-0347-11](#)
- EMHS [IHPT Roles, Responsibilities and Escalation flow chart](#)

7. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.15 High-risk medicines
- Standard 6 – Action 6.07 and 6.08 Clinical handover
- Standard 6 – Action 6.03 Partnering with consumers in Clinical communication
- Clinicians use organisational processes from the Partnering with Consumers Standard to effectively communicate with patients, carers and families during high-risk situations to:
- Standard 6 – Action 6.09 Communicating critical information

- Standard 8 – 8.04 Recognising acute deterioration
- Standard 8 – Action 8.06 Escalating care

9. References

1. UpToDate: Initial evaluation and management of suspected acute coronary syndrome (myocardial infarction, unstable angina) in the emergency department (2019).
2. Australian Commission on Safety and Quality in Health Care. Indicator Specification: Acute Coronary Syndromes Clinical Care Standard. Sydney: ACSQHC, 2019
3. National Heart Foundation of Australia & Cardiac Society of Australia and New Zealand: Australian Clinical Guidelines for the Management of Acute Coronary Syndromes 2016
4. NSW Chest Pain Pathway – Chest pain evaluation (2019)
5. QLD Government: Possible Cardiac Chest Pain Clinical pathway (2019)

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
Acute Coronary Syndrome (ACS)	An umbrella term used to describe a spectrum of symptoms ACS comprises of distinct diagnostic groups: ST elevation myocardial infarction (STEMI), non-ST elevation myocardial infarction (NSTEMI) and Unstable Angina (UA) and Non-ST segment elevation acute coronary syndrome (NSTEMACS)
Acute Myocardial Infarction (AMI)	Acute myocardial infarction occurs when an atherosclerotic plaque ruptures, triggering a complex cascade of events that result in thrombi partially or completely occluding the coronary vasculature, compromised coronary blood flow and subsequent injury and/or death of myocardial tissue
Unstable Angina (UA)	Angina occurs when a plaque or blood clot results in narrowing of the blood vessel. This may happen as a precursor to a myocardial infarction or remain stable for a long time.

11. Policy owner (relating to these procedures)

Nursing Coordinator

Enquiries relating to this procedure may be directed to:

Title: Nursing Coordinator
 Department: Hospital wide
 Email: jane.husain@health.wa.gov.au

Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
Version 2.0	08/09/2022	08/09/2025	Scheduled review

12. Authorisation

This procedure has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Dr Alison Parr, Director Clinical Services
Authorisation date	08/09/2022
Published date	08/09/2022

Appendix 1: Chest pain tool L.O.C.A.T.E.

Assess nature and location of suspected or confirmed ACS chest pain using the L.O.C.A.T.E. criteria.

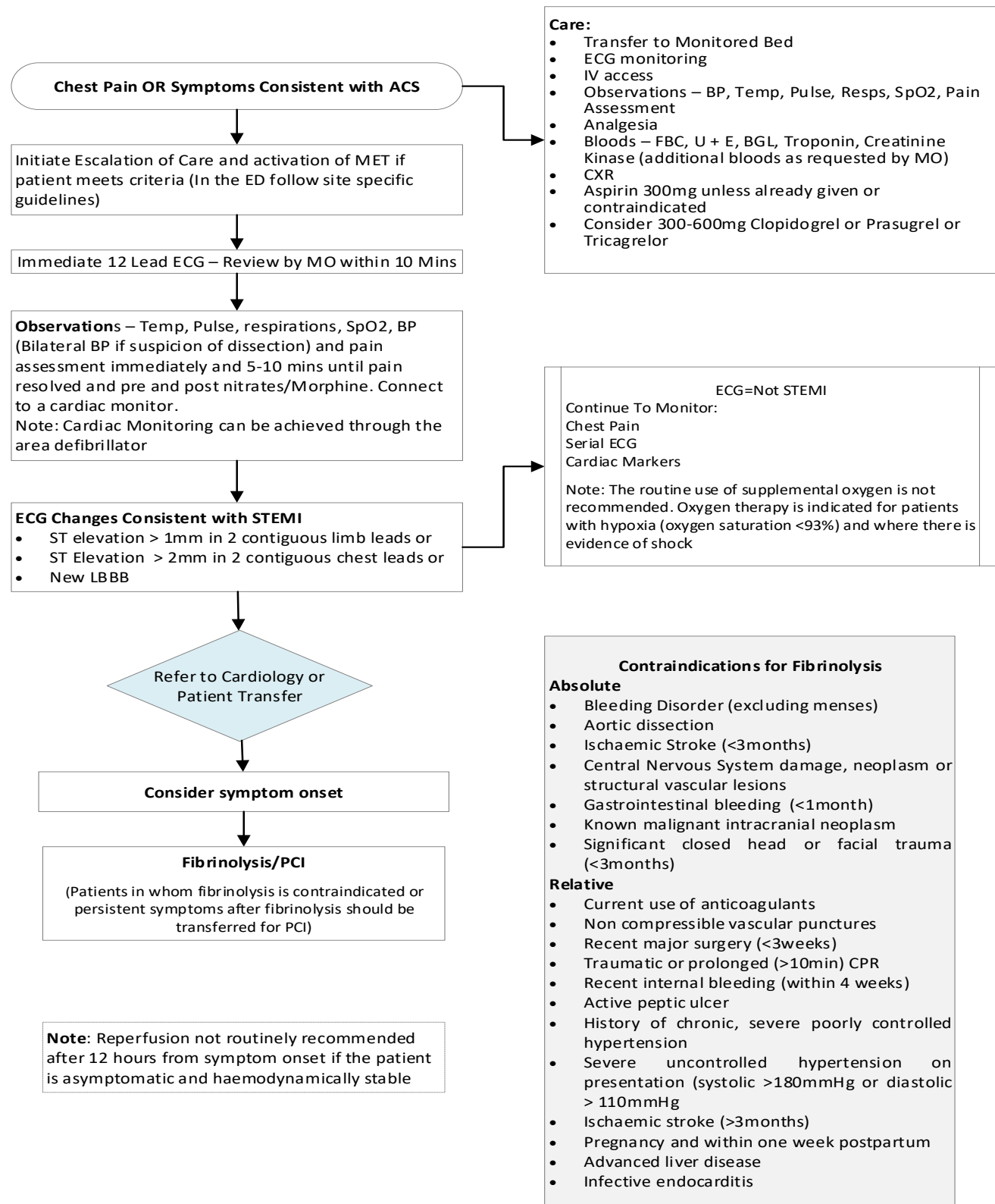
Assess for the most life-threatening cause before considering conditions with less risk.

L ocation	- Exact location e.g. chest, neck, back or arm - Does the pain radiate and if so where? - Use anatomical landmarks to describe the exact location e.g. 5th intercostal space mid clavicular line
O nset and D uration	- What was the exact time of the chest pain onset? - Was it gradual or acute? - When does it occur? - What was the patient doing at the time? - How long did it last? - Does it change or move? - Does it increase or decrease in intensity?
C haracter	- Ask the character of the pain. Note they may need help with descriptive words e.g. sharp, stabbing, dull, aching, burning, pressure, heaviness, tightness, cramping - Document the patients pain score (0-10) both now and also when the pain started
A ssociated S ymptoms	- Ask and observe for symptoms e.g. nausea, vomiting, sweating, dizziness, oedema, shortness of breath, palpitations, fatigue. - Check for symptoms of muscular-skeletal pain - Check for vague symptoms, especially in females such as sleep pattern changes
T reatment	What has the patient tried to do to relieve the pain both now and in the past? E.g. - GTN spray / tablets - Antacids, H2 antagonist - Rest stopped activity - Stretching/meditation/relaxation exercise - Breathing techniques - Heat or ice pack Did it work?
E liminate /Aggravate	- What eliminates the pain? E.g. Rest, medication, stopping the activity or any of the previous treatments (above) - What aggravates or instigates the pain? E.g. Movement, deep respirations or coughing, eating - Does food alleviate or aggravate the pain? - Is the pain changed by leaning forward?

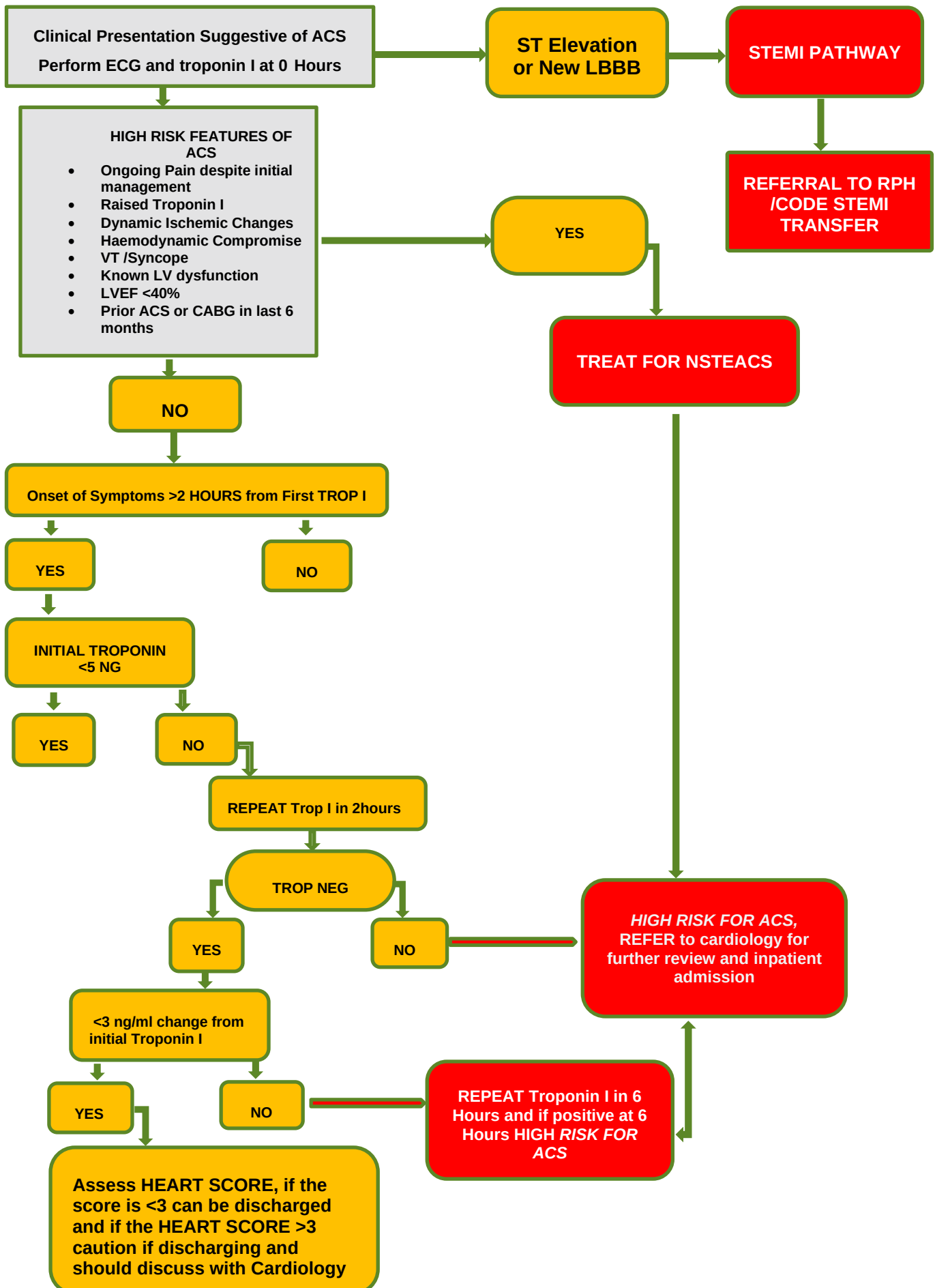
Appendix 2: Stratify ACS Risk table

Stratify ACS Risk		
High Risk Admit or Transfer	Intermediate Risk Restratify	Low Risk Discharge
☐ ACS symptoms are repetitive of prolonged (>10 min) & still present. ☐ Syncope ☐ History of chronic left ventricular systolic dysfunction (especially if known LVEF < 40%) OR current clinical evidence of LVF. ☐ Previous PCI/CABG < 6 months ☐ Diabetes + typical ACS symptoms ☐ Chronic renal failure + typical ACS symptoms ☐ Haemodynamic compromise (sustained SBP <90 mmHg and / or new onset mitral regurgitation) ☐ Elevated Troponin (consider haemolysis, renal failure)	☐ ACS symptoms within 48 hours that occurred at rest, or were repetitive or prolonged (but currently resolved) ☐ Previous PCI/CABG > 6 months ☐ Known coronary artery disease. Especially if prior AMI or known coronary lesion >50% stenosis ☐ Two or more risk factors of: Hypertension, family history, active smoking or hyperlipidaemia ☐ Chronic renal failure (esp. if known GFR < 60 mL/min) + atypical ACS symptoms ☐ Diabetes + atypical ACS symptoms ☐ Age > 65 years	☐ Presentation with clinical features consistent with ACS without intermediate risk or high risk features
☐ Persistent or dynamic ECG changes of: • ST depression >5mm or • New T wave inversion >2mm ☐ Transient ST elevation (>0.5mm) in more than two contiguous leads ☐ Sustained VT	☐ ECG is not normal and has not changed from previous pain free ECG but does not contain high risk changes	☐ ECG normal or unchanged from previous pain free ECG

Appendix 3: Acute Coronary Syndrome (ACS) assessment / treatment algorithm



Appendix 4: Troponin assessment and time pathway



Appendix 5: HEART SCORE**HEART SCORE**

Heart score is a scoring system to assess the risk for developing a major acute coronary event in patients above the age of 21 presenting with Chest pain. It should not be used for acute STEMI or hemodynamically unstable patients. The HEART is an acronym used for **H**istory, **E**CG, **A**ge, **R**isk **F**actors and Trend in **T**roponin

History	- Highly suspicious - Moderately suspicious - Low Suspicion	2 points 1 point 0 point
ECG	- ST depression - Non-Specific changes (e.g. T wave changes) - Nil Changes in ECG	2 points 1 point 0 Point
AGE	- Age >65 - Age 45-64 - Age <45	2 points 1 point 0 point
RISK FACTORS*	>3 risk factors >1 - <3 Risk factors - No Risk factors	2 points 1 point 0 point
TROPONIN	3 values of greater than the Normal limit 1-<3 values of greater than Normal limit - Normal Troponin	2 1 0

- RISK OF MAJOR CARDIAC EVENT (MACE) IS LOW IF THE SCORE IS 0-3**
- RISK OF MAJOR CARDIAC EVENT (MACE) IS MODERATE IF THE SCORE IS 4-6**
- RISK OF MAJOR CARDIAC EVENT (MACE) IS HIGH IF THE SCORE IS 7-10**

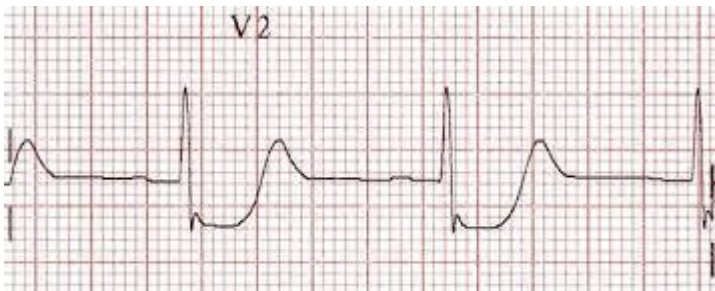
**Risk factors include HTN, DM, Atherosclerosis, Smoking (less than 3 months of cessation), family history, CABG, History of PCI, TIA, peripheral vascular disease,*

Appendix 5: STEMI EQUIVALENTS

- DEWINTER SIGN



- POSTERIOR WALL MI



- NEW LEFT BUNDLE BRANCH BLOCK

Left bundle branch block characteristics

